How to use Lapack on Nuvolos

1. Start application `Visual Studio Code 1.10.2.1`.

2. Grab the bottom line of the screen and pull it up to display the terminal.

3. Check the content of `/opt/conda/lib`:
   Use command: `ls *blas*`, the output is
   `libblas.so libblas.so.3 libcblas.so libcblas.so.3 libopenblasp-r0.3.28.so libopenblas.so.0`
   Use command: `ls *lapack*`, the output is
   `liblapack.so liblapack.so.3`.

4. Check the content of `/opt/conda/include`:
   Use commands: `ls *blas*` and `ls *lapack*`
   You can see that the `blas` and `lapack headers` are not installed.

5. Install missing applications:
   ```
   conda install openblas-devel -c anaconda
   Confirm changes:  Y
   ```

6. Check that the headers are now present in `/opt/conda/include`:
   ```
   ls *blas*
   cblas.h  f77blas.h  openblas_config.h
   ls *lapack*
   lapacke_config.h  lapacke.h  lapacke_mangling.h  lapacke_utils.h  lapack.h
   ```

7. Add useful commands and the paths at the end of the `.bashrc` file in the connection directory:
   ```
   alias cp="cp -i"
   alias rm="rm -i"
   alias mv="mv -i"
   export PATH=$PATH:.
   #Libraries for loading (LD)
   export LDFLAGS="-L/opt/conda/lib"
   #Headers for C pre-processor (CPP):
   export CPPFLAGS="-I/opt/conda/include"
   ```

8. Add `gfortran` to install `gcc` and `gfortran` in the `Conda` environment:
   Use command: `conda install gfortran`

9. In directory `Lapack`, create the script `compil.sh` for compilation and linking:
   `gcc $LDFLAGS $CPPFLAGS -o Lapack_C_double.exe Lapack_C_double.c -llapack -lblas -lcblas`

10. Compile the program `Lapack_C_double.c` with Blas and Lapack:
    `bash compil.sh`